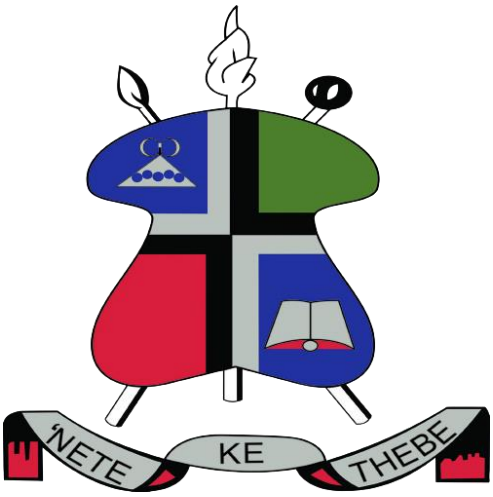


National University of Lesotho
Faculty of Science and Technology
Department of Mathematics and Computer Science

Project Management Plan

Learning Management System

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Course	CS4433 – Software Engineering
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Version	1.0

1. Feasibility Study:

Technical feasibility:

The proposed LMS is technically feasible because the required technology stack is defined as C++17, SQLite, CMake, Git and an automated testing framework. The application is intended to operate on a local network. The LMS on a single computer will not be suitable because the system should respond quickly even with all 800 students on it, hence must support multiple access. Also, on the LAN the system will reduce dependence on Internet connectivity. This addresses the IT officer's requirement that the system work offline. However, the school's old computers and the requirement to support data for up to 800 students introduce performance and compatibility considerations. These can be addressed through performance testing and testing on school hardware. Password protection and secure storage must also be incorporated because the system handles student and assessment information. Assessment: We can build this system. We just need to test its speed and safety first.

Operational feasibility:

The system is operationally feasible because its functions correspond closely to the existing responsibilities of the school's students, lecturers, Heads of Department and administrator. Role based access will ensure that each user performs only permitted activities. However, the principal's concern that teachers are not IT experts means usability is important. Training and clear user documentation will therefore be required. The system's offline operation is also appropriate given the school's unreliable Internet connection. Assessment: Operationally feasible, provided usability and training requirements are addressed.

Economic feasibility:

The case study does not provide a specific project budget, so a quantitative ROI (Return on Investment) calculation cannot be justified at this stage. However, the project uses an existing C++/SQLite technology stack and is designed to run on a single local network, potentially reducing infrastructure and hosting costs. The system can provide benefits through centralized assignment and assessment management, reduced manual administration and improved traceability of mark changes. Costs will include development, testing, training, maintenance and potentially upgrading old school computers. Overall, the project appears economically feasible. Assessment: Economically feasible, as long as we check and confirm that we actually have the budget and tools needed to start it.

Legal feasibility:

The system is legally feasible provided data-protection and education requirements are incorporated into its design and operation. The LMS will process student personal information and assessment records, so appropriate access controls and password protection are necessary. The system must not store passwords in plain text and must maintain an audit trail of mark changes. Copyright and ownership of uploaded learning materials should also be considered. The project brief specifically identifies data protection, security of assessment records, accessibility and fairness of late-penalty rules as professional, legal and ethical considerations. Assessment: Legally feasible, provided applicable legal and security requirements are implemented.

Schedule feasibility:

The project is schedule feasible because the assignment provides a seven-week schedule covering planning, requirements, design, implementation, verification and validation, deployment and maintenance. However, the schedule is constrained because implementation cannot begin until the design has passed, while testing overlaps the latter part of implementation. Any delay in requirements or design could therefore affect implementation and subsequent testing. Milestone monitoring and early identification of risks will be necessary to remain within the seven-week period. Assessment: Schedule-feasible, but high schedule pressure exists.

Feasibility Summary Table:

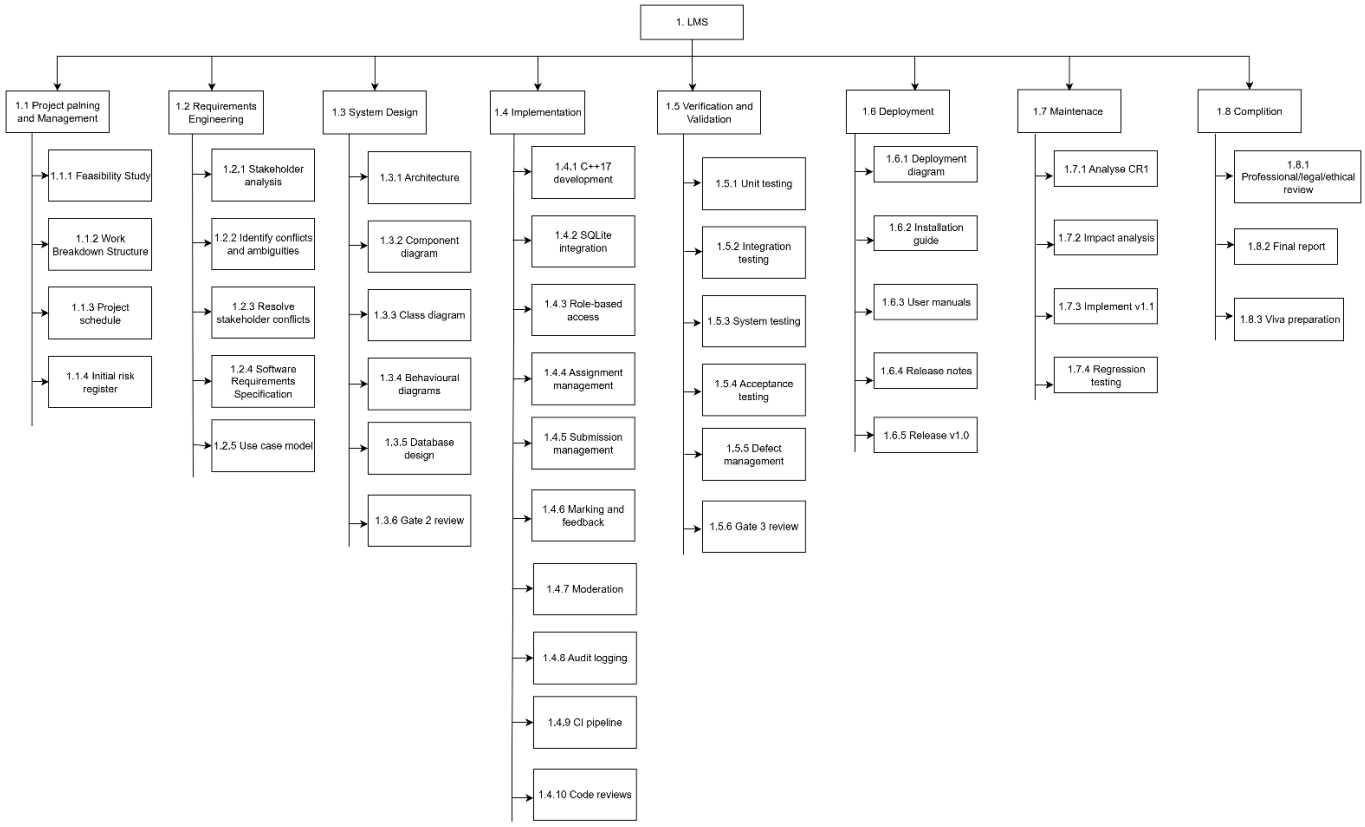
Feasibility	**Finding**	**Main concern**	**Assessment**
Technical	C++17/SQLite/local deployment is achievable	Old computers and 800-user dataset	Feasible with performance/security testing
Operational	Functions align with school roles	Teachers may require training	Feasible
Economic	Existing/local technology limits infrastructure needs	Budget not provided	Feasible subject to resource confirmation
Legal	Can implement security, access control and auditing	Student/assessment data protection	Feasible subject to resource confirmation
Schedule	Seven-week Waterfall plan is defined	Tight dependencies and approval checkpoints	Feasible subject to resource confirmation

Recommendation:

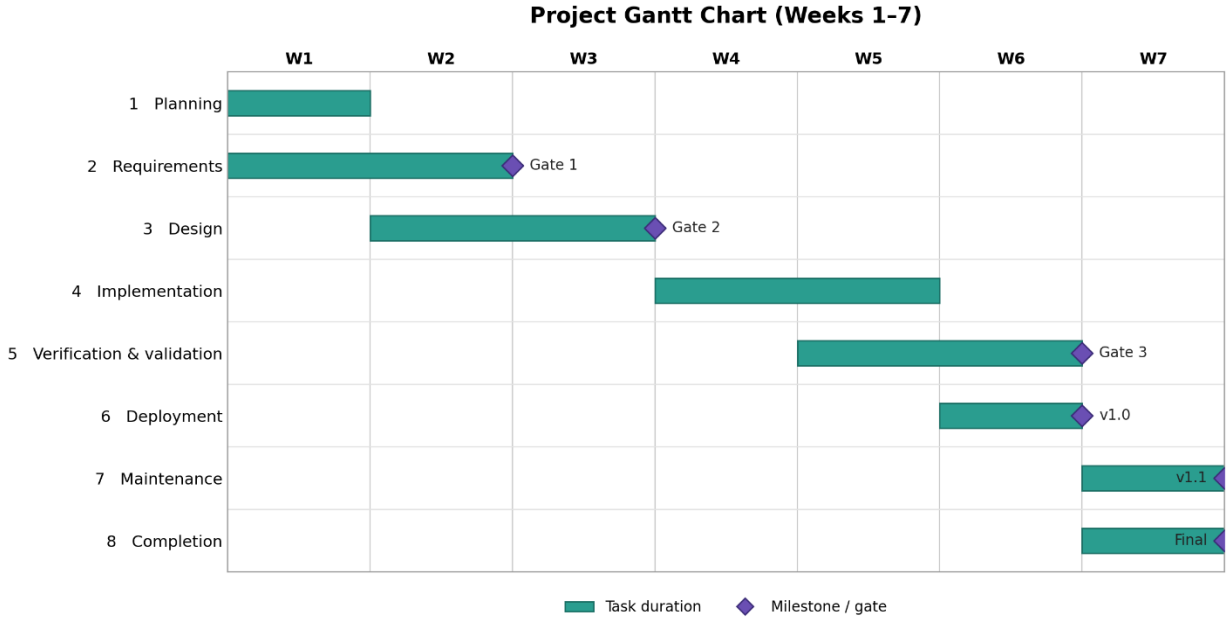
The feasibility assessment indicates that the moderated LMS is technically, operationally, economically, legally and schedule feasible. The strongest constraints are the school's old computers, unreliable Internet, the seven-week development period, and the need to protect student and assessment information. These issues can be addressed through offline/local operation, performance testing, security controls, phased reviews and active risk management. The project should therefore proceed to requirements engineering, provided the required resources are confirmed and the identified constraints are monitored throughout the project.

2. Work breakdown and schedule:

Work Breakdown Structure:



Schedule:



Critical path and Dependencies:

Planning → Requirements → Gate 1 → Design → Gate 2 → Implementation → Verification & Validation → Gate 3 → Deployment → Maintenance → Completion

3. Risk management:

The scoring scale below is used for project risk management:

Likelihood:

Score	Meaning
1	Very unlikely
2	Unlikely
3	Possible
4	Likely
5	Almost certain

Impact:

Score	Meaning
1	Very low
2	Low
3	Moderate
4	High
5	Very high

Risk Exposure:

Exposure	Rating
1 - 4	Low
5 – 9	Moderate
10 – 16	High
17 - 25	Very High

$$\textbf{Risk Exposure} = \textbf{Likelihood} \times \textbf{Impact}$$

Risk Register:

ID	Risk	Category	Likelihood	Impact	Exposure	Mitigation	Owner
R1	Old school computers may cause compatibility or performance problems.	Technical	3	4	12	Test the application on representative school hardware and optimize resource-intensive operations.	Technical Lead
R2	The system may not perform well with records for all 800 students.	Technical	3	4	12	Create a test dataset representing 800 students and do performance testing before deployment.	Technical Lead
R3	Security weaknesses could expose student information, passwords or assessment records.	Technical	3	5	15	Use hashed passwords, role-based access control, parameterized queries and security testing.	Security/Technical Lead
R4	Conflicting stakeholder requirements may cause requirements changes and rework.	Project	4	4	16	Conduct stakeholder analysis, resolve conflicts during requirements engineering.	Project Manager
R5	Delays in requirements or design may reduce the time available for implementation and testing.	Project	4	5	20	Monitor milestones weekly, resolve issues early and maintain buffer time for delays.	Project Manager
R6	The design may fail Gate 2, preventing implementation from starting on time.	Project	3	5	15	Conduct design reviews before Gate 2 and verify architecture, UML and database design against requirements.	Project Manager / Technical Lead
R7	Teachers may have difficulty using the system because they are not IT experts.	People	3	4	12	Keep workflows simple and provide training and user documentation.	Project Manager / Training Lead
R8	Stakeholders may disagree about assessment rules, particularly marks, extensions, resubmissions and late penalties.	People/Project	4	4	16	Resolve business-rule conflicts during requirements engineering and obtain stakeholder approval before design.	Requirements Lead

The risk register will be reviewed at each phase gate. Risks whose likelihood or impact has changed will be re-scored, completed risks will be closed, and newly identified risks will be added. Gate 1 follows requirements engineering, Gate 2 follows system design, and Gate 3 follows verification and validation.

Gate 1 update table:

Risk	**Initial Exposure**	**Gate 1 Exposure**	**Status**	**Reason for change / action**
R1 Old computer compatibility	12	12	Open	Hardware issue remains until implementation testing.
R2 800-student performance	12	12	Open	Performance cannot be fully verified until implementation.
R3 Security vulnerability	15	15	Open	Security design and implementation are still required.
R4 Stakeholder conflicts	16	8	Reduced	Conflicts resolved and requirements baselined.
R5 Schedule delay	20	16	Reduced	Requirements completed, but remaining schedule is still tight.
R6 Gate 2 failure	15	15	Open	Design has not yet been completed/reviewed.
R7 Teacher usability	12	12	Open	Usability will be addressed through design and training.
R8 Assessment-rule disagreement	16	8	Reduced	Assessment rules clarified in the SRS.

Gate 2 update table:

Risk	Gate 1	Gate 2	Status	Reason
R1 Old computer compatibility	12	8	Reduced	Design has considered resource constraints; implementation testing remains.
R2 800-student performance	12	12	Open	Requires implementation/performance testing.
R3 Security vulnerability	15	10	Reduced	Security controls incorporated into design.
R4 Stakeholder conflicts	8	4	Low	Requirements and business rules have been baselined.
R5 Schedule delay	16	16	Open	Implementation and testing remain under tight time constraints.
R6 Gate 2 failure	15	0 / Closed	Closed	Gate 2 successfully passed.
R7 Teacher usability	12	9	Reduced	Usability requirements incorporated into design.
R8 Assessment-rule disagreement	8	4	Low	Rules incorporated into design and requirements.

Gate 3 update table:

Risk	Gate 2	Gate 3	Status	Reason
R1 Old computer compatibility	8	4	Low	Testing confirms acceptable operation on representative hardware.
R2 800-student performance	12	4	Low	Performance testing completed successfully.
R3 Security vulnerability	10	5	Reduced	Security testing completed and significant defects addressed.
R4 Stakeholder conflicts	4	2	Low	Requirements remain stable and acceptance testing confirms expected behavior.
R5 Schedule delay	16	8	Reduced	Major implementation and testing activities completed.
R6 Gate 2 failure	Closed	Closed	Closed	Design gate was successfully passed.
R7 Teacher usability	9	4	Low	User acceptance testing and training materials address usability concerns.
R8 Assessment-rule disagreement	4	2	Low	Assessment rules verified through acceptance testing.